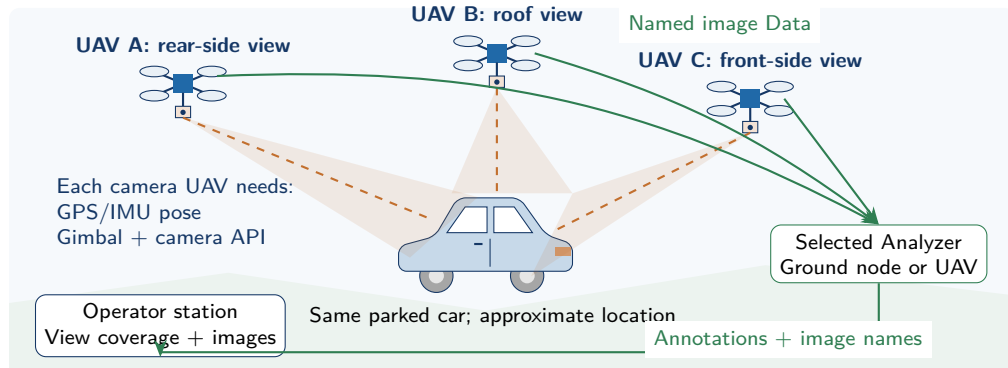


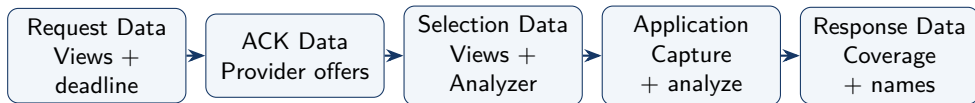
A Controlled Task: Collect Useful Views of One Car

Remote visual check: an operator requests rear, roof and front views of one parked car before a deadline.



Evidence: Proposed field test, not a completed flight experiment. Orange: views; green: named-data retrieval, not flight paths or unsolicited push.

Why This Is a Useful NDNF Test Case



- ▶ **Availability:** select camera and compute Providers at request time.
- ▶ **Dependencies:** retrieve selected views by name; report coverage and missing views.
- ▶ **Access rights:** authorize services and protect images across participating devices.

Key point

Compare against one UAV and ground compute; measure whether collaboration is worth its cost.

Evidence: Proposed mapping: NDNF handles invocation; the application validates capture and same-car association.

Three Gaps Between Simulation and the Field

Gap	Simplified example	Real deployment needs
Position / pointing	Known coordinates; assigned views	GPS + IMU, camera/gimbal calibration, coordinate frames, synchronized pose and exposure
Useful images	Car assumed visible	Real control/capture APIs, feedback, blur/occlusion checks and bounded retries
Multi-view result	Images supplied to a detector	Same-car association, requested-view coverage and independent image labels

Key point

Start with verified, complementary images. Damage diagnosis and 3-D reconstruction require separate algorithms and validation.

Evidence: Candidate tools: MAVSDK control/telemetry; OpenCV calibration/geometry. Hardware support and integration remain to be verified.

Candidate Hardware for the First Camera Bench Test

SIYI A8 mini

SIYI



- ▶ Camera + three-axis gimbal in one unit.
- ▶ Ethernet UDP / UART control; Ethernet video.

Gremsy Pixy U



- ▶ Gimbal only; add a compatible camera.
- ▶ SDK / MAVLink control; verify camera support.

First bench test: command an angle → read actual attitude → capture the parked car.

Evidence: Product images: SIYI official store and Gremsy official site. Images not to scale. Candidate hardware; verify power, payload and mounting before selection.